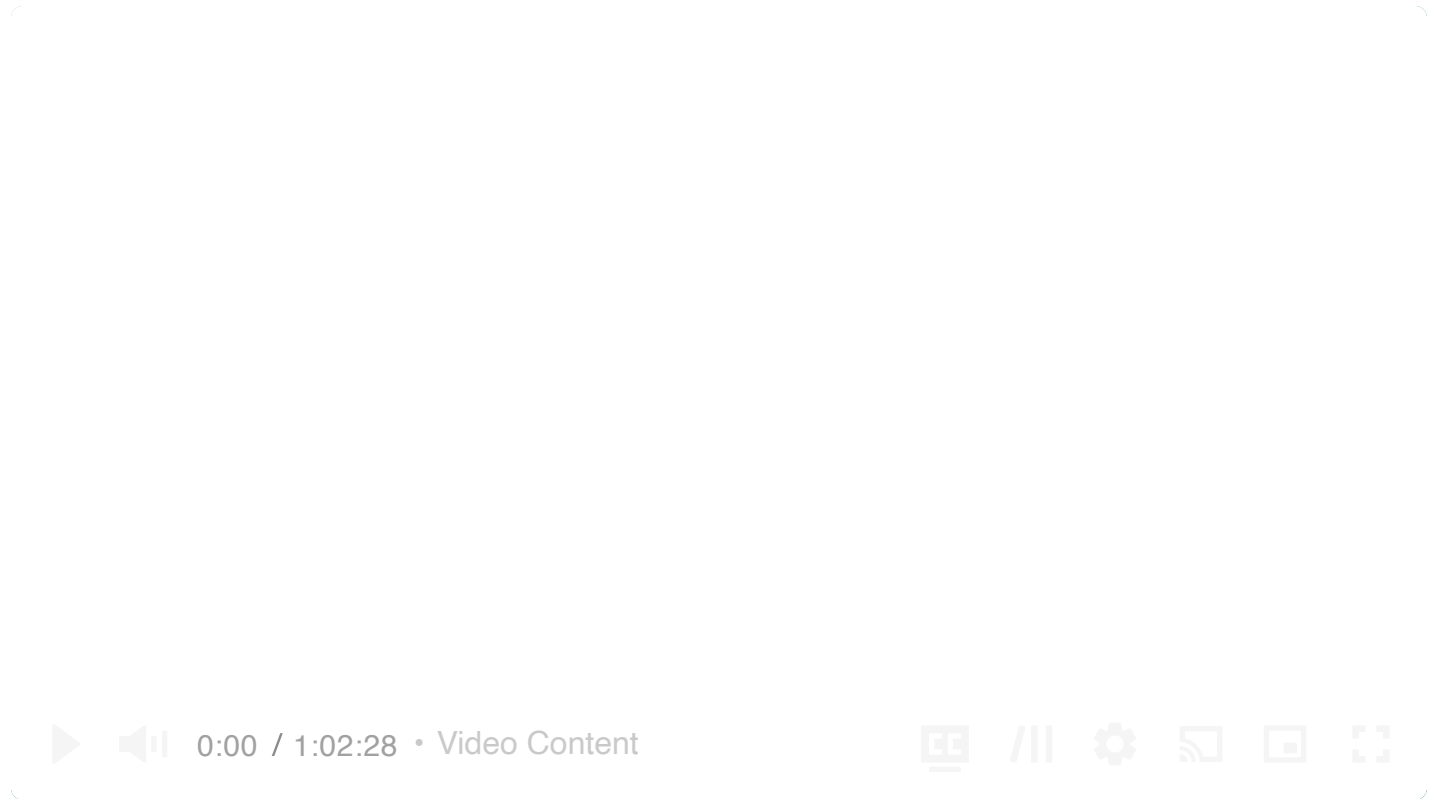


Common Problems



Price Tracking Service 🎧

By Evan King · Published Jun 4, 2025 · medium · Asked at:



Understanding the Problem

What is CamelCamelCamel?

CamelCamelCamel is a price tracking service that monitors Amazon product prices over time and alerts users when prices drop below their specified thresholds. It also has a popular Chrome extension with 1 million active users that displays price history directly on Amazon product pages, allowing for one-click subscription to price drop notifications without needing to leave the Amazon product page.

Functional Requirements

Core Requirements

1. Users should be able to view price history for Amazon products (via website or Chrome extension)
2. Users should be able to subscribe to price drop notifications with thresholds (via website or Chrome extension)



CamelCamelCamel has a popular Chrome extension with 1 million active users that displays price history directly on Amazon product pages, allowing for one-click subscription to price drop notifications without needing to leave the Amazon product page.

Below the line (out of scope):

- Search and discover products on the platform
- Price comparison across multiple retailers
- Product reviews and ratings integration

Non-Functional Requirements

The scale and performance requirements for CamelCamelCamel are driven by Amazon's massive product catalog and the need for timely price notifications.

Core Requirements

1. The system should prioritize availability over consistency (eventual consistency acceptable)
2. The system should handle 500 million Amazon products at scale
3. The system should provide price history queries with < 500ms latency
4. The system should deliver price drop notifications within 1 hour of price change

Below the line (out of scope):

- Strong consistency for price data
- Real-time price updates (sub-minute)



We're building a system that must be "polite" to Amazon while providing valuable price tracking to millions of users. This creates interesting technical challenges around data collection, storage efficiency, and notification delivery that we'll address in our deep dives.

Here is how your requirements might look on the whiteboard:

Functional Requirements

- view price history for all amazon products
- subscribe to price drops based on price threshold

Non-Functional Requirements

- availability >> consistency
- scale to 500m Amazon products
- low latency price history queries (<500ms)
- timely price drop notifications (<1 hr from change)

CamelCamelCamel Requirements

The Set Up

Planning the Approach

We'll follow the Hello Interview framework by building our design sequentially through each functional requirement, then use our non-functional requirements to guide our deep dive discussions on scaling challenges.

Defining the Core Entities

I like to begin with a broad overview of the primary entities we'll need, without getting bogged down in all the details yet. We can always add more complexity as we iterate on the design.



Just make sure that you let your interviewer know your plan so you're on the same page. I'll often explain that I'm going to start with just a simple list, but as we get to the high-level design, I'll document the data model more thoroughly.

To satisfy our key functional requirements, we'll need the following entities:

1. **Product:** Represents an Amazon product we're tracking, including its current price and metadata needed for price history display and notifications
2. **User:** Represents a person using our service, containing their contact information and preferences for receiving price drop notifications

3. **Subscription:** Links users to products they want to monitor, specifying the price threshold that triggers notifications
4. **Price:** Time-series data capturing price changes over time for each product, enabling us to generate historical price charts

In the actual interview, this can be as simple as a short list like this. Just make sure you talk through the entities with your interviewer to ensure you are on the same page.

Core Entities

- Product
- User
- Subscription
- Price

CamelCamelCamel Entities

The API

The API is the main way users will interact with our price tracking service, both through our website and the Chrome extension. Let's design endpoints that directly support our two core functional requirements.

```
// Retrieve historical price data for charts and displays
GET /products/{product_id}/price?period=30d&granularity=daily -> PriceHistory[]
```



We include `product_id` in the URL path rather than the request body for better caching and REST compliance. The `granularity` parameter allows us to return daily averages for long periods or hourly data for recent timeframes.

```
// Subscribe to price drop notifications with threshold
POST /subscriptions
{
  product_id,
```

```
price_threshold,  
notification_type,  
}  
-> 200
```

Data Flow

Before diving into the technical design, let's understand how data flows through our system. The data flow shows how information moves from initial collection to final user output.

Understanding this flow early in our design process serves multiple purposes. First, it helps ensure we're aligned with our interviewer on the core functionality before getting into implementation details. Second, it provides a clear roadmap that will guide our high-level design decisions. Finally, it allows us to identify potential bottlenecks or issues before we commit to specific architectural choices.

In this case, the "hidden" requirement is that we need to be able to get the data in the first place!

1. Web crawlers and Chrome extension collect current prices from Amazon product pages
2. Price data is validated and stored in our price database
3. Price changes trigger events for notification processing
4. User receives email notification when price drops below threshold



Note that this is simple, we will improve upon as we go, but it's important to start simple and build up from there.

High-Level Design

We'll build our design incrementally, starting with the most fundamental requirement and adding complexity as we address each successive need. This ensures we deliver a working system first, then layer on additional capabilities.



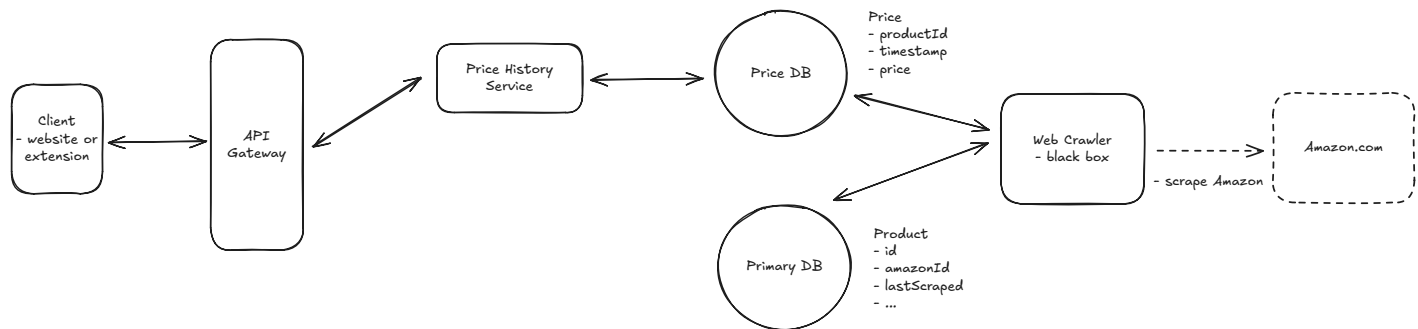
Amazon is not friendly to price tracking services. They don't provide an API, actively discourage scraping, and implement rate limiting (typically 1 request per second per IP

address). We must design our system to work within these constraints while remaining respectful of Amazon's terms of service.

1) Users should be able to view price history for Amazon products (via website or Chrome extension)

Users want to see how prices have changed over time for specific Amazon products. This serves as the foundation of our entire system and breaks down into two distinct challenges: acquiring price data from Amazon and serving historical price queries efficiently.

For the first iteration, we'll establish the basic data collection and storage infrastructure. Our architecture needs a web scraping system to gather price data, a storage layer for historical prices, and an API service to serve price history to users.



CamelCamelCamel High Level Design - Price History

Here's what we're building:

1. **Client**: Our client is either the website or the Chrome extension. In either case, they're how the user views price history graphs.
2. **API Gateway**: Handles authentication, rate limiting, and routes requests to appropriate services
3. **Price History Service**: Manages price data retrieval and coordinates with the database
4. **Web Crawler Service**: Scrapes Amazon product pages to collect current prices
5. **Price Database**: Stores both current product information and historical price data
6. **Primary Database**: Stores all other tables aside from price like **Users** , **Products** , and **Subscriptions** .

We choose to separate the Price History Service from the crawler because they have different scaling characteristics. The history service needs to handle many concurrent read requests from users viewing charts, while the crawler operates on a scheduled basis with consistent write patterns. This separation also allows us to optimize each service independently.

As for the databases, we'll make a strategic separation based on scale and access patterns. Our Primary Database will store **Users** , **Products** , and **Subscriptions** together since these have similar operational characteristics - they're relatively small and have traditional CRUD access patterns. However, we'll put our Price Database completely separate because price history data has fundamentally different requirements: it will grow to billions of rows, is append-only, requires time-series optimizations, and can tolerate eventual consistency. This separation allows us to optimize each database independently and scale them based on their specific needs.

We'll choose to "black box" the web crawler service for now. We can assume it periodically scrapes Amazon product pages and updates the **Price** table. Of course, we'll go into more detail about just how this might be accomplished in our deep dives.

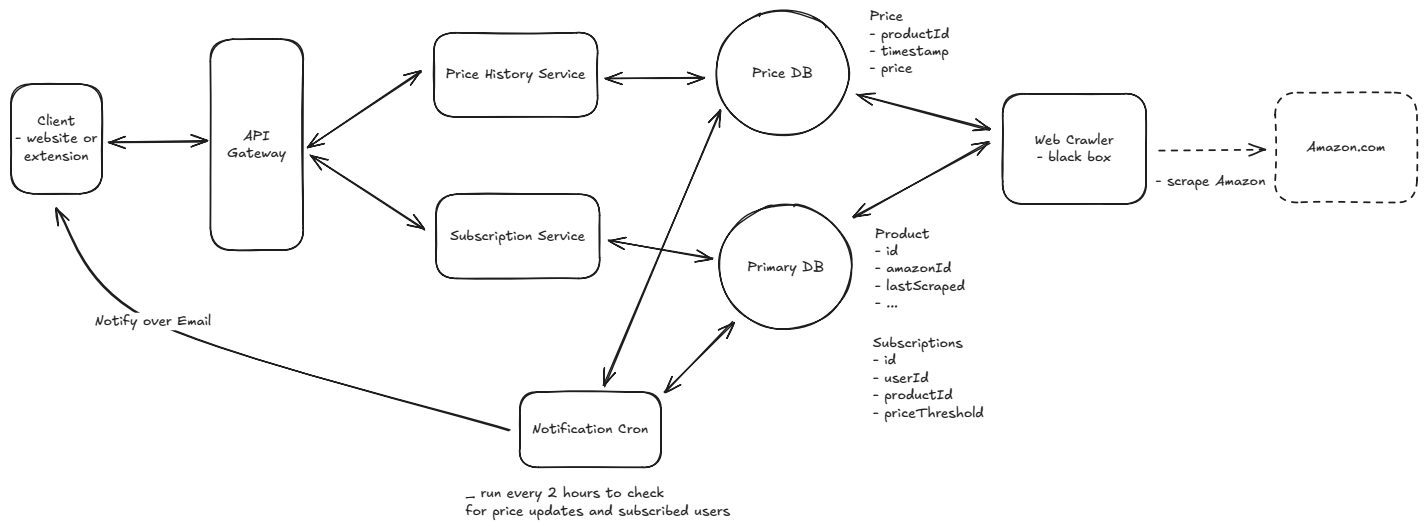
When a user requests price history through our website or Chrome extension:

1. Client sends GET request to `/products/{product_id}/price?period=30d`
2. API Gateway authenticates the request and routes to Price History Service
3. Price History Service queries the **Price** table filtered by `product_id` and time range
4. Historical prices are returned as JSON, ready for chart rendering
5. Chrome extension or website displays the price chart to the user

2) Users should be able to subscribe to price drop notifications with thresholds (via website or Chrome extension)

Building on our price history foundation, users now want to receive notifications when product prices drop below their specified thresholds. This introduces new requirements for user management, subscription storage, and notification delivery.

We'll add new components while leveraging our existing infrastructure. For our initial implementation, we can start with a dead simple polling approach that gets the job done, knowing we can optimize this later in our deep dives.



CamelCamelCamel High Level Design - Subscriptions

Our enhanced architecture includes:

1. **Subscription Service:** Manages user subscriptions and price threshold settings
2. **Notification Cron Job:** Periodically checks for price changes and sends email notifications

We separate the Subscription Service from the Price History Service because they serve different user actions and have different data access patterns. The Subscription Service handles user account operations (create, update, delete subscriptions), while the Price History Service focuses on read-heavy price queries.

For our initial notification system, we have a straightforward batch processing approach:

1. Cron job runs every 2 hours to check for notifications
2. Job queries **Price** table for any price changes in the last 2 hours
3. For each price change, query **Subscriptions** table from the Primary Database to find users with thresholds $\geq$ new_price
4. Send email notifications for any triggered subscriptions
5. Mark notifications as sent to avoid duplicates



This polling approach works but has obvious limitations. Users might wait up to 2 hours for price drop notifications, and we're doing expensive database scans every few hours. We'll explore much better real-time approaches in our deep dives that provide sub-minute notification delivery.

When a user subscribes to price alerts:

1. User submits subscription via website or Chrome extension
2. API Gateway routes POST request to Subscription Service
3. Subscription Service creates record in Subscriptions table (user_id, product_id, price_threshold)
4. Service returns confirmation to user

This simple approach makes sure we have a working subscription system that satisfies our functional requirements. Users get price drop alerts (just not immediately), and our system remains easy to understand and debug. The periodic nature also prevents notification processing from interfering with our core price tracking functionality.

Potential Deep Dives

Time for the fun part. We'll take our existing, simple design and layer on complexity via our deep dives. When we're done, we'll have a system that satisfies all of our functional and non-functional requirements.

As one of my favorite sports journalists Fabrizio Romano would say, "Here we go!"

1) How do we efficiently discover and track 500 million Amazon products?

Up until now, we've been "black boxing" our web crawler, simply assuming it somehow gets price data from Amazon and updates our database. But when we confront the reality of tracking 500 million products while respecting Amazon's rate limiting constraints, we realize this is actually one of our biggest technical challenges.

We're solving two distinct but related problems here.

The first challenge is product discovery: we need to find all 500 million existing products when we launch, plus discover the approximately 3,000 new products Amazon adds daily.

The second challenge is price monitoring: we need to efficiently update prices for millions of known products while prioritizing which ones to check most frequently.

Both challenges face the same fundamental constraint of Amazon's rate limiting, but understanding them separately helps us design more effective solutions.

Bad Solution: Naive Web Crawling Approach



Good Solution: Prioritized Crawling Based on User Interest



Great Solution: Chrome Extension + Selective Crawling



This hybrid approach solves our scale challenge by turning our biggest constraint (the need to monitor millions of products) into our competitive advantage through crowdsourced data collection. Instead of fighting Amazon's rate limits with massive crawler infrastructure, we leverage user behavior to naturally prioritize and collect data for the products that matter most.



If you're thinking to yourself, "How the hell would I know this? I know nothing about Chrome Extensions!" don't sweat it. When I ask this in an interview, I usually lead with, "Your extension is installed on millions of browsers actively viewing product pages. Can you make use of that somehow?"

2) How do we handle potentially malicious price updates from Chrome extension users?

The Chrome extension approach from our previous deep dive creates a powerful data collection advantage, but it introduces a critical reliability challenge. With 1 million users potentially submitting price data, we must assume that some percentage will be malicious actors, mistaken users, or systems experiencing technical issues.

Consider this scenario: A malicious user reports that the latest iPhone costs \$0.01, triggering price drop notifications to thousands of subscribers. Users rush to Amazon only to find the normal price, damaging our credibility and user trust. Our notification system's effectiveness depends entirely on data accuracy, making this challenge essential to solve properly.

So what should we do?

Good Solution: Consensus Based Validation



Great Solution: Trust-But-Verify with Priority Verification



This trust-but-verify approach strikes a really nice balance, providing immediate user value while maintaining the data integrity essential for long-term trust.

3) How do we efficiently process price changes and notify subscribed users?

Right now, we're using a straightforward cron job that runs every 2 hours to scan for price changes and send notifications. While this might work for basic functionality, it fails to meet our non-functional requirement of delivering notifications within 1 hour of price changes. More critically, the polling approach creates massive database load as we scale to millions of price updates and subscriptions.

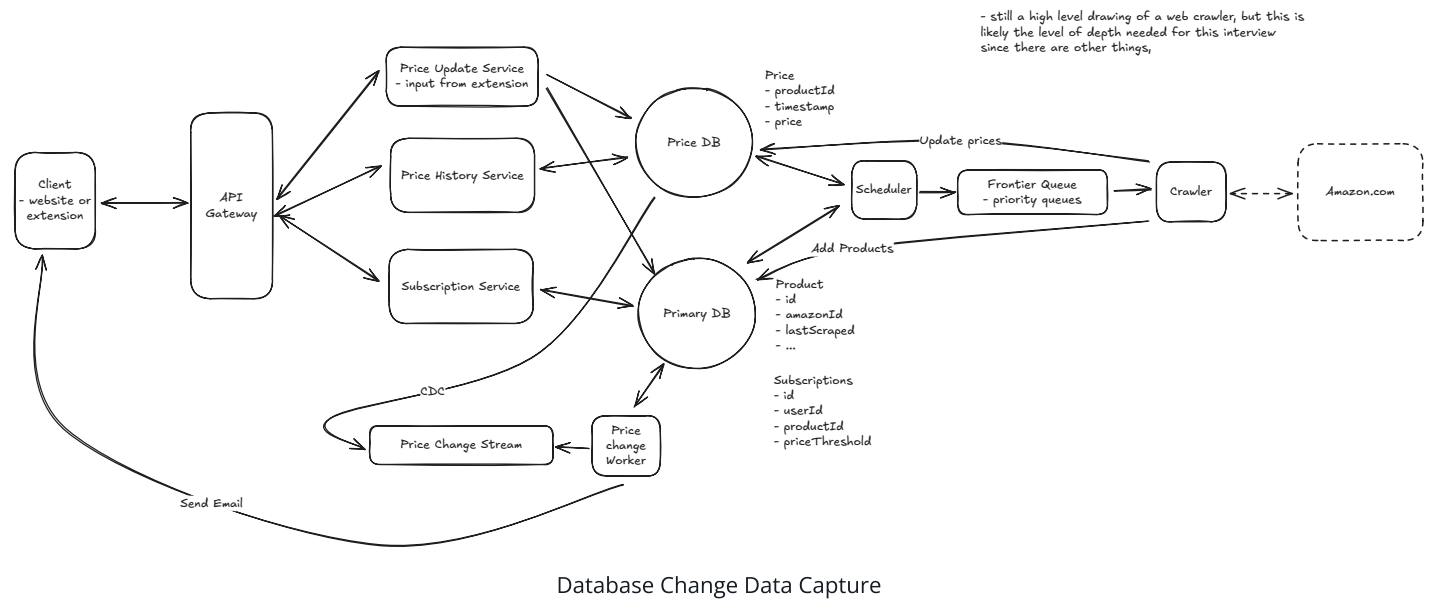
The notification system's effectiveness directly impacts user satisfaction and retention. Users expect timely alerts for price drops, especially for competitive deals that might sell out quickly. A 2-hour delay often means missing time-sensitive opportunities, making our service less valuable than competitors with faster notification systems.

The fundamental problem with our current polling approach is that we're asking the wrong question at the wrong time. Instead of asking "what changed in the last 2 hours?" every 2 hours, we should be asking "who cares about this change?" immediately when a change happens.

Moving to event-driven notifications is straightforward. When a price changes, we immediately know which users need alerts without expensive database scans. We have two solid ways to implement this, each with different trade-offs.

The first option is database change data capture (CDC), where we configure our database to automatically publish events whenever price data changes. When our crawlers or extension data processors insert new price records, database triggers fire and send events to Kafka (or

similar) containing the product ID, old price, and new price. Our notification service subscribes to these events and immediately queries the subscriptions table to find affected users.



This is clean because price changes automatically trigger notifications without any application-level coordination. We don't need to remember to publish events in our application code since the database handles it for us. The data flow becomes completely automatic.

The second option is dual writes, where our price collection services write to both the database and publish events to Kafka simultaneously. When crawler data or extension updates come in, the same service that writes to the price database also publishes structured events to our notification stream.

The dual-write approach lets us be smarter about which changes trigger notifications. We can filter out tiny price fluctuations that wouldn't interest users or batch multiple rapid changes before publishing events. We also avoid the overhead of database triggers on every single price update.

Regardless of which option we choose, our notification consumers work the same way. They read price change events from Kafka and immediately query the subscriptions table to find users whose thresholds have been triggered. For each matching subscription, we send an email notification through our existing notification service.

The beauty is that we only process actual price changes rather than expensive table scans of millions of records. Most products don't change price very frequently, so we're talking about a manageable number of events even at scale. When a popular product's price drops, we run

one query to find all subscribers with thresholds above the new price and send those notifications right away.

4) How do we serve fast price history queries for chart generation?

Looking at our current data storage approach, we're keeping price history in a straightforward PostgreSQL table with basic indexing on (product_id, timestamp). This works for initial functionality, but fails to meet our < 500ms latency requirement when serving price charts for popular products with extensive history data.

Consider the scale challenge that comes from aggregating price data for popular products. Popular products might have price data points every few hours for several years, resulting in many thousands of records per product. When users request a 2-year price chart, we must aggregate this data into appropriate time buckets (daily or weekly averages) and return it quickly enough for smooth user experience. Raw database queries struggle with this aggregation workload.

Good Solution: Scheduled Pre-Aggregation with Cron Jobs



Great Solution: TimescaleDB for Real-Time Price Analytics



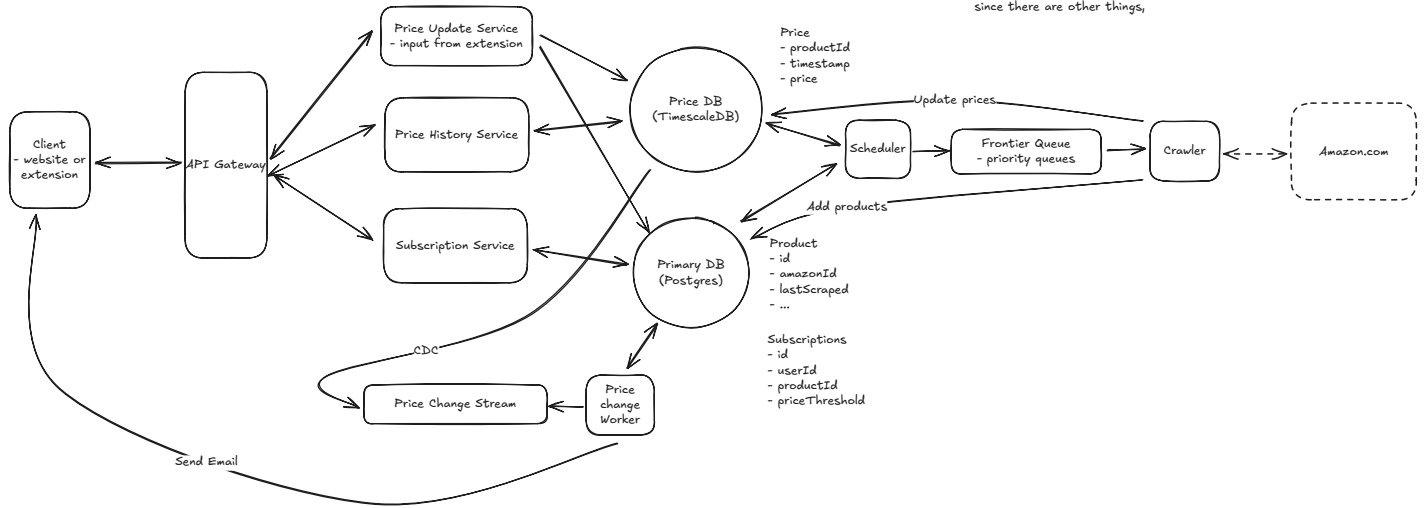
The TimescaleDB approach provides the performance and flexibility needed for production-scale price chart serving while maintaining the real-time responsiveness users expect. By choosing the right tool for analytical workloads, we achieve both simplicity and performance without complex caching or pre-aggregation strategies.

Final Design

Taking a step back, we've designed a scales system that can intelligently collect price data from Amazon, validate it, and notify users when prices drop all while serving price history charts to users in real-time. Not bad!

As for the final diagram, I'll admit the drawing got a little out of hand with all the crossing arrows. But we end up with a final design that looks something like this!

- still a high level drawing of a web crawler, but this is likely the level of depth needed for this interview since there are other things,



Final Design

What is Expected at Each Level?

So, what am I looking for at each level when I ask candidates to design a price tracking system like CamelCamelCamel?

Mid-level

At mid-level, I'm looking for candidates who can create a working high-level design that addresses the core functional requirements - price history display and notification subscriptions. They should understand the basic challenge of collecting price data from Amazon and propose a simple web scraping approach, even if it's not scalable. I expect them to design a straightforward database schema with products, users, subscriptions, and price history. While they might initially suggest polling every product frequently, with some guidance they should recognize the scale limitations and understand why prioritization might be necessary.

Senior

For senior candidates, I expect you to quickly identify that data collection is the core technical challenge and drive the conversation toward solutions like the Chrome extension approach. You should understand the trade-offs between different crawling strategies and understand that crawling 500M products will require either immense resources or endless time. I'm looking for solid understanding of notification system architectures, you should be able to explain why

polling doesn't scale and propose event-driven alternatives using message queues. Ideally you go deep on at least of couple of deep dives, even if not the ones I have above.

Staff+

For staff candidates, I'm evaluating your ability to see the bigger picture and balance technical elegance with operational reality. You should proactively recognize that the Chrome extension isn't just a user feature, but that it can be used to solve the fundamental data collection scaling problem. I expect you to discuss system evolution thoughtfully: how do we start simple but design for the scale we'll eventually need? You should understand the business implications of technical choices, for example, why data validation affects user trust and retention, not just technical correctness. Strong candidates often surface concerns I haven't asked about, like handling Amazon's anti-scraping measures, managing extension user privacy, or planning for what happens when Amazon changes their page structure. You demonstrate systems thinking by considering how different components affect each other and proposing solutions that prioritize long-term maintainability alongside immediate functionality.

Test Your Knowledge

Take a quick 15 question quiz to test what you've learned and identify gaps.

 [Start Quiz](#)